



Noakhali Science & Technology University

PROJECT TITLE: QUICKBITE-ONLINE FOOD ORDERING SYSTEM

Introduction and requirement analysis

Software Engineering and Information System Design Lab

Course Code: CSTE3310

Submitted to

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1. Introduction (About the Project)

QuickBite: Online Food Ordering System

QuickBite is a web-based food ordering platform designed to simplify the process of ordering meals online. The system allows customers to browse restaurants, search food items, add items to a cart, place orders, make online payments, and track deliveries in real time.

Restaurants can manage their menus, process customer orders, and update order status. Administrators supervise the overall system, manage users, restaurants, and generate analytical reports.

The system aims to provide a convenient, efficient, and reliable food ordering experience while improving restaurant management and customer satisfaction.

2(a). Project Scope

The scope of QuickBite includes:

Customer Side

- ✚ User Registration
- ✚ Login Authentication
- ✚ Restaurant Browsing
- ✚ Food Searching
- ✚ Cart Management
- ✚ Order Placement
- ✚ Online Payment
- ✚ Order Tracking
- ✚ Ratings and Feedback

Restaurant Side

- ✚ Restaurant Registration
- ✚ Menu Management
- ✚ Order Acceptance
- ✚ Order Processing
- ✚ Updating Delivery Status

- ✚ Sales Monitoring

Delivery Side

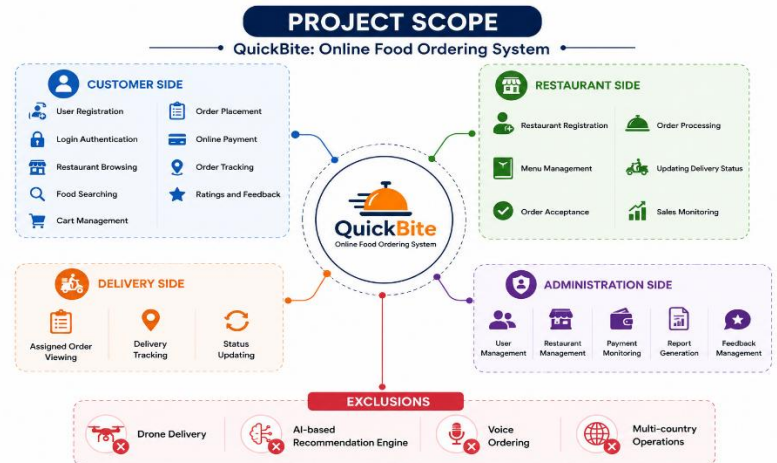
- ✚ Assigned Order Viewing
- ✚ Delivery Tracking
- ✚ Status Updating

Administration Side

- ✚ User Management
- ✚ Restaurant Management
- ✚ Payment Monitoring
- ✚ Report Generation
- ✚ Feedback Management

Exclusions

- ✚ Drone Delivery
- ✚ AI-based Recommendation Engine
- ✚ Voice Ordering
- ✚ Multi-country Operations



2(b). Objectives and Goals

Objectives

- ✚ Develop an online food ordering platform.
- ✚ Reduce customer waiting time.
- ✚ Enable secure payment processing.
- ✚ Improve restaurant order management.
- ✚ Facilitate real-time order tracking.
- ✚ Increase operational efficiency.

Goals

- ✚ User-friendly interface.
- ✚ Fast order processing.
- ✚ Secure data handling.

- ✚ High availability.
- ✚ Scalability for future expansion.
- ✚ Better customer satisfaction.

3. Defining Requirements

3(a). Defining All Requirements

QuickBite should provide functionalities that support customers, restaurants, delivery personnel, and administrators.

The system must allow:

- ✚ Customer registration
- ✚ Authentication
- ✚ Menu browsing
- ✚ Searching foods
- ✚ Cart operations
- ✚ Order management
- ✚ Payment processing
- ✚ Delivery tracking
- ✚ Restaurant management
- ✚ Feedback handling
- ✚ Reporting facilities



3(b). Functional Requirements

FR1

The system shall allow customers to register.

FR2

The system shall authenticate users.

FR3

Customers shall browse restaurants.

FR4

Customers shall search food items.

FR5

Customers shall add items to cart.

FR6

Customers shall modify cart contents.

FR7

Customers shall place orders.

FR8

The system shall process payments.

FR9

Restaurants shall manage menus.

FR10

Restaurants shall accept or reject orders.

FR11

Restaurants shall update order status.

FR12

Delivery personnel shall update delivery progress.

FR13

Customers shall track orders.

FR14

Customers shall provide ratings.

FR15

Administrators shall generate reports.



3(c). Non-Functional Requirements

Performance

System response time should be less than 3 seconds.

Availability

System availability should exceed 99%.

Reliability

Transactions must be processed accurately.

Security

Passwords must be encrypted.

Payment transactions should be secured.

Scalability

The system should support increasing users.

Maintainability

Easy modification and updating capability.

Usability

Simple and intuitive user interface.

Compatibility

Accessible through desktop and mobile browsers.

4. Requirement Analysis

(a) Scenario-Based Models

i. Use Case Diagram

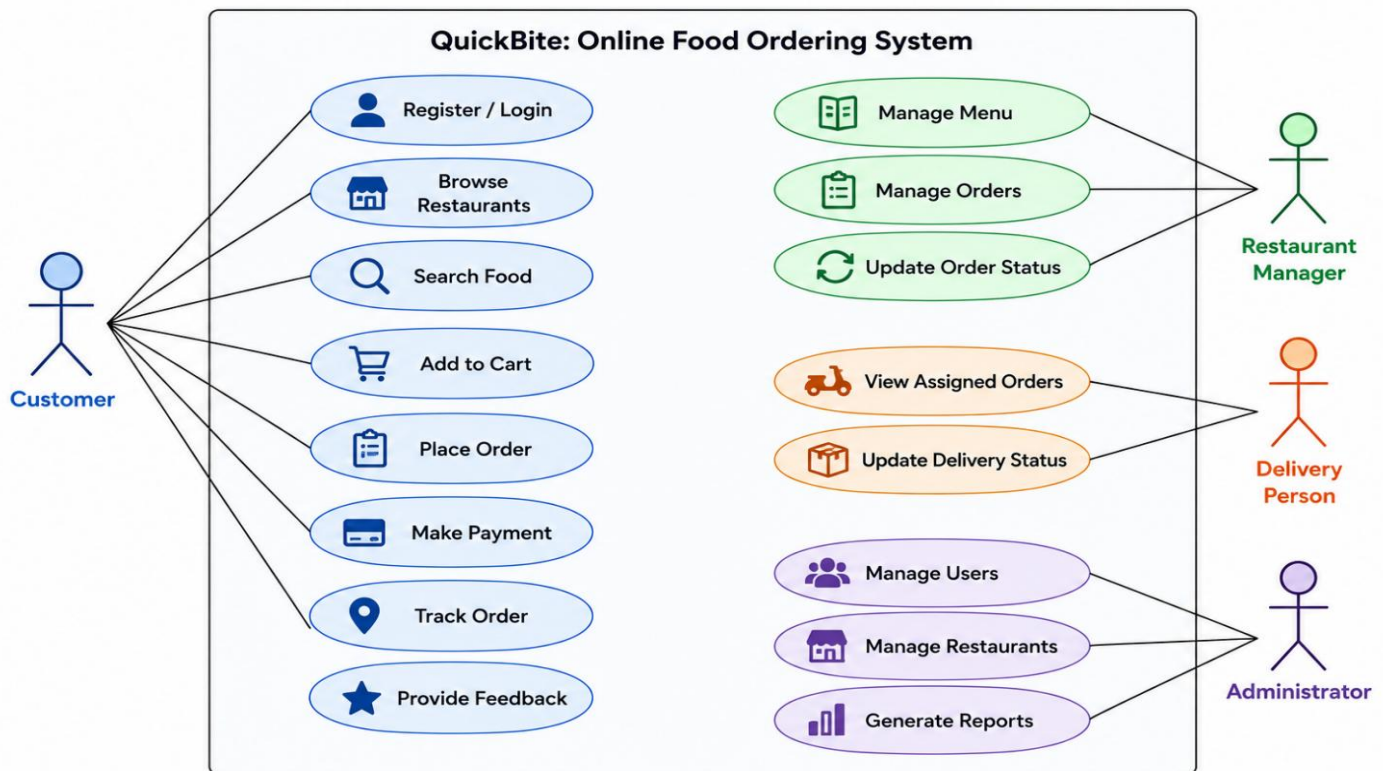
Primary Actors

-  Customer
 -  Restaurant Manager
 -  Delivery Person
 -  Administrator
-

Secondary Actors

- ✚ Payment Gateway
- ✚ Email Service
- ✚ Notification Service

Use Case Diagram – QuickBite: Online Food Ordering System



ii. Use Case Description

3 major use cases.

Use Case Description – 1

Use case: Place Order

Actors: Customer (Initiator)

Type: Primary and Essential

Description:

The customer logs into the QuickBite system and browses available restaurants and food items. The customer selects desired items and adds them to the shopping cart. The customer may continue browsing and adding more items. After selecting all items, the customer proceeds to checkout, confirms the delivery address, and places the order. The system generates an order ID and forwards the order to the restaurant.

Cross Ref.:

Requirements FR2, FR3, FR4, FR5, FR7

Use-Cases:

Login, Browse Restaurants, Search Food, Add to Cart, Checkout

Use Case Description – 2

Use case: Make Payment

Actors: Customer (Initiator)

Payment Gateway (Secondary Actor)

Type: Primary and Essential

Description:

The customer selects a payment method during checkout. If online payment is selected, the system redirects the customer to the payment gateway. The payment gateway verifies the transaction information and processes the payment. Upon successful payment, the system updates the payment status and confirms the order. If the payment fails, the customer is notified and may retry the transaction.

Cross Ref.:

Requirements FR8, FR13

Use-Cases:

Place Order, Verify Payment, Generate Receipt

Use Case Description – 3

Use case: Manage Menu

Actors: Restaurant Manager (Initiator)

Type: Primary and Essential

Description:

The restaurant manager logs into the QuickBite system and accesses the menu management module. The manager can add new food items, update existing items, modify prices, or remove unavailable items. The system validates the information and updates the food database accordingly.

Cross Ref.:

Requirements FR9, FR10

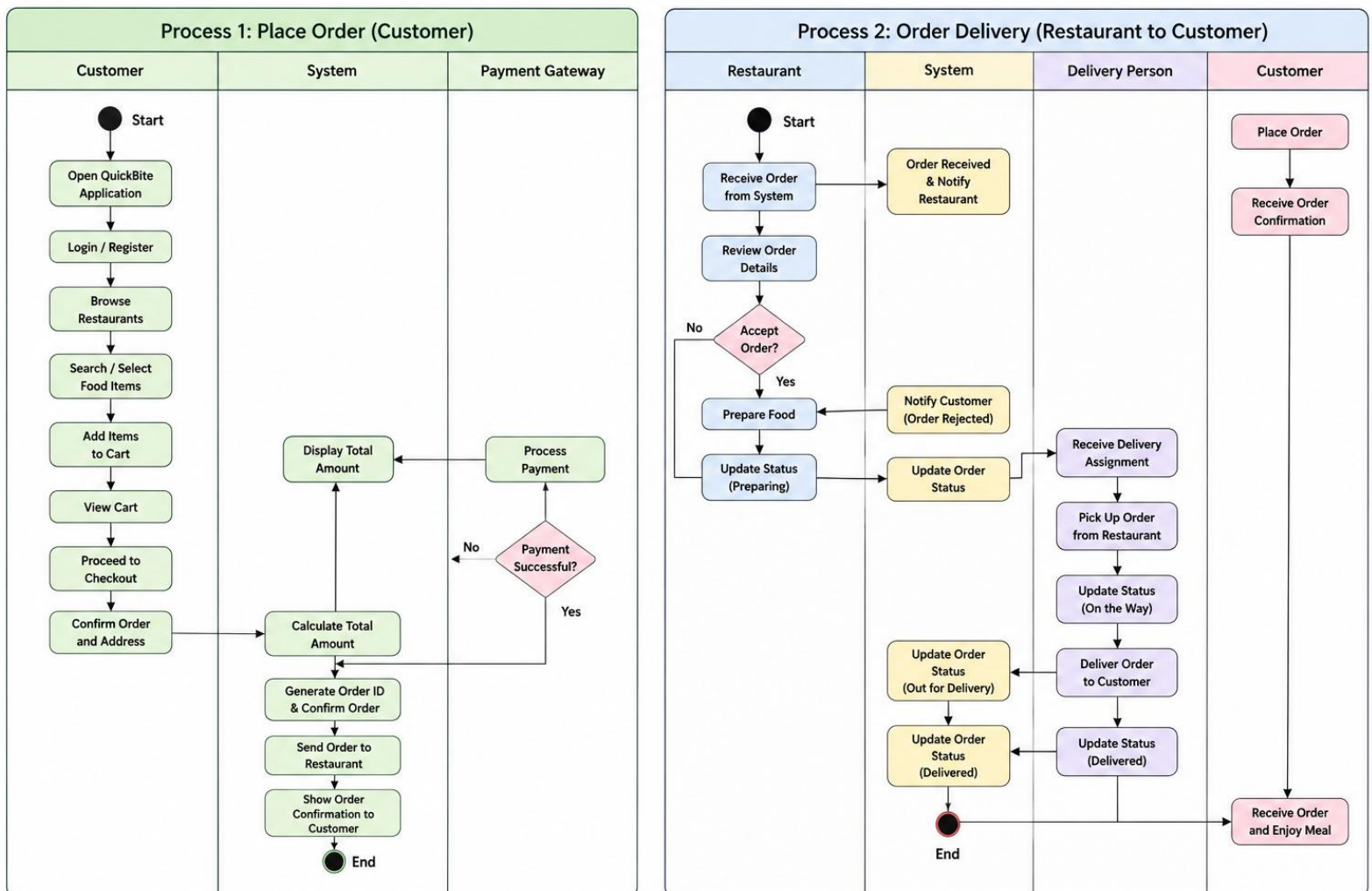
Use-Cases:

Login, Add Food Item, Update Food Item, Delete Food Item

4(b). Behavioral Models

i. Activity Diagram (2 major processes)

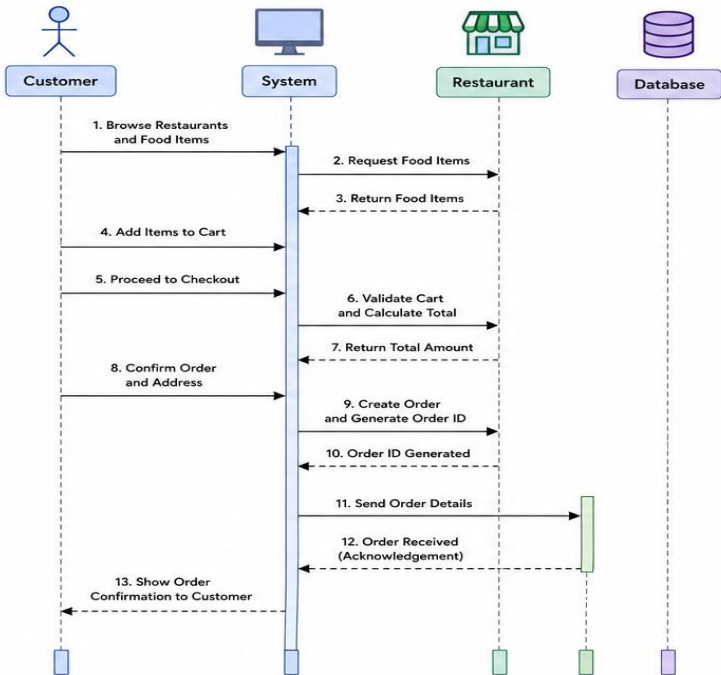
Activity Diagram – QuickBite: Online Food Ordering System



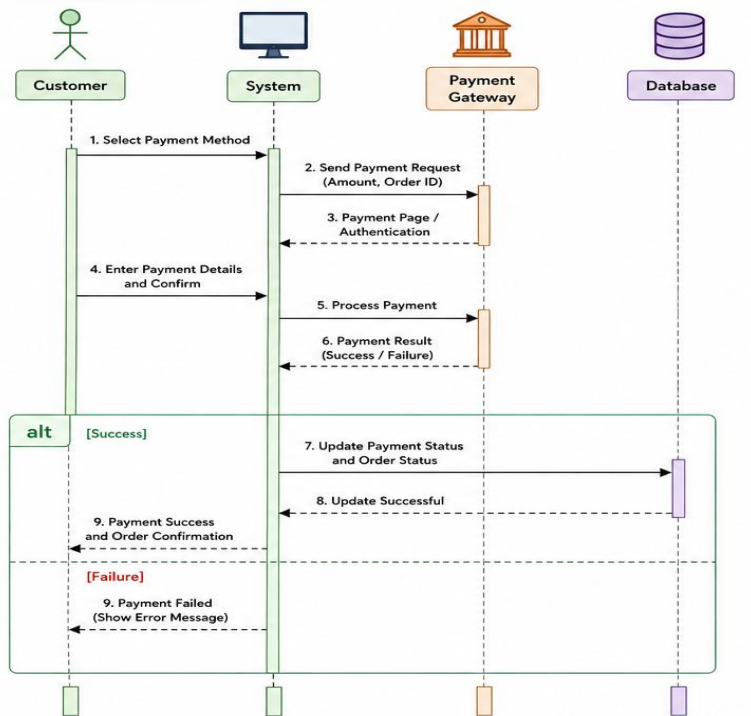
ii) Sequence Diagram (2 major processes)

Sequence Diagrams – QuickBite: Online Food Ordering System

Sequence Diagram 1: Place Order



Sequence Diagram 2: Payment Process

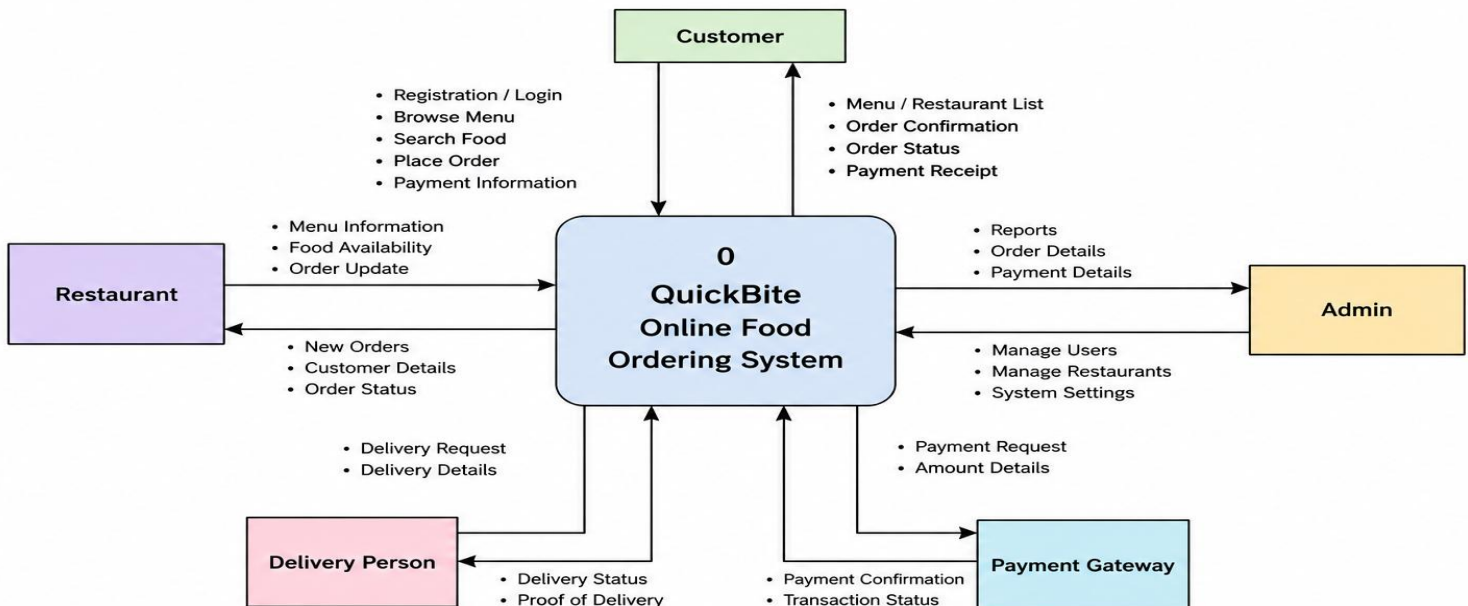


5. Requirement analysis

(a) Flow models –

i. Level 0 DFD :

DFD Level 0 Diagram – QuickBite: Online Food Ordering System



Legend

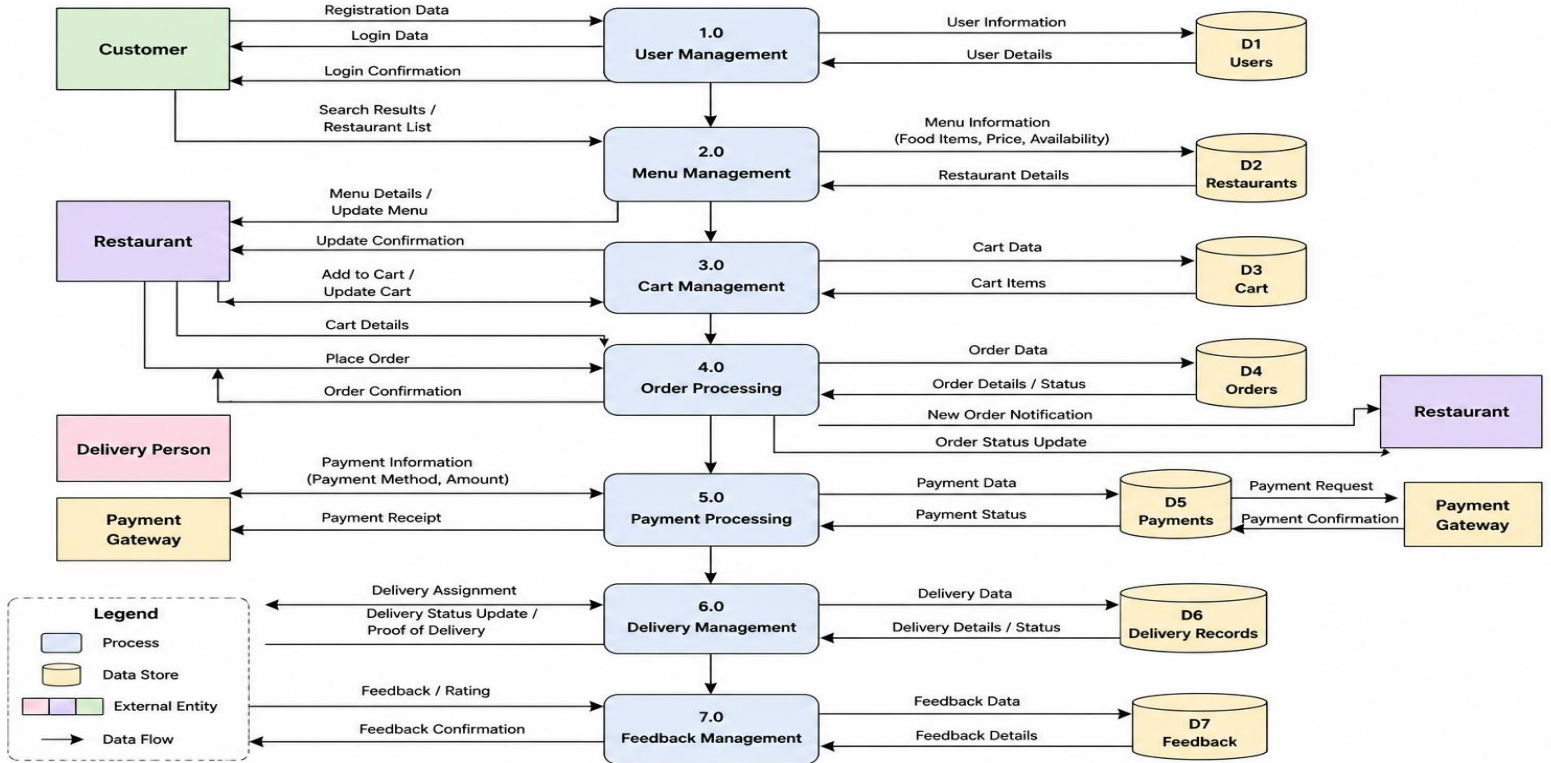


Note:

This is a Level 0 DFD (Context Diagram) showing the QuickBite system as a single process and its interactions with external entities.

ii. Level 1 DFD :

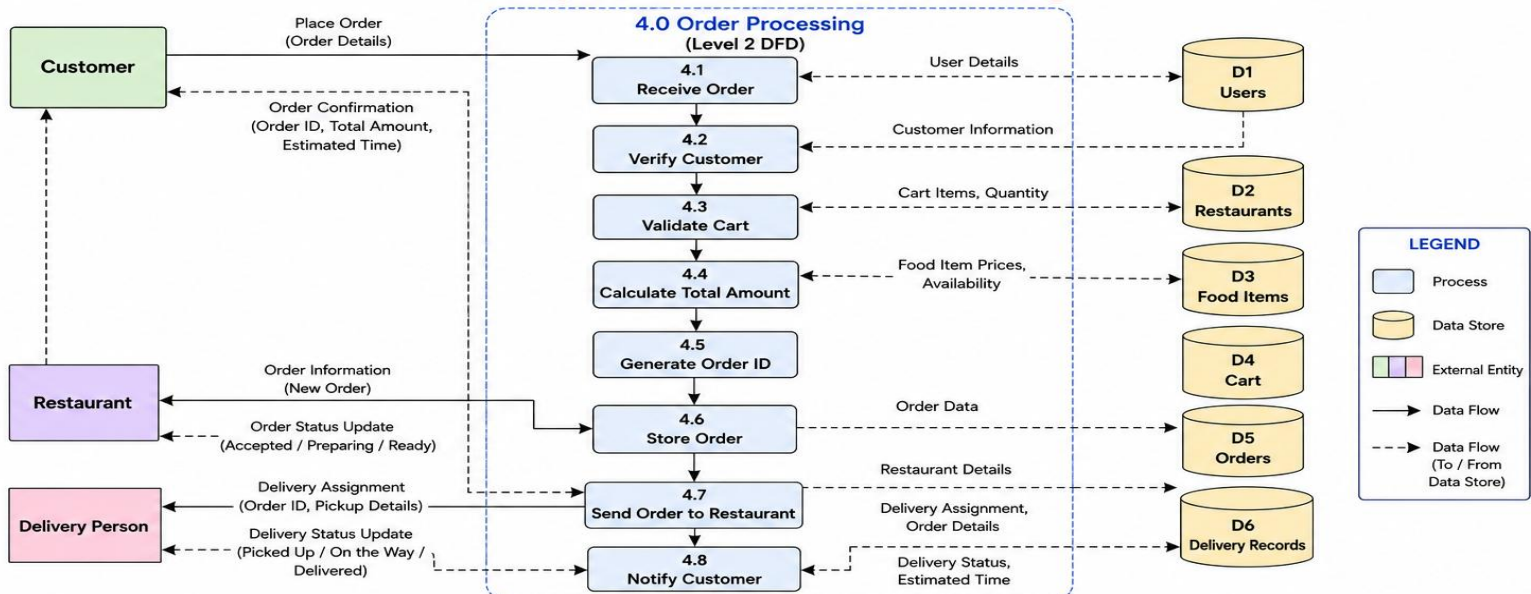
DFD Level 1 Diagram – QuickBite: Online Food Ordering System



iii. Level 2 DFD :

DFD Level 2 Diagram – QuickBite: Online Food Ordering System

(Decomposition of Process 4.0 Order Processing)



PROCESS 4.0 DESCRIPTION

This process handles the complete flow of an order from the time it is placed by the customer to the time it is delivered. It includes order validation, price calculation, order storage, sending the order to the restaurant, assigning delivery and finally notifying the customer.

LEVEL 2 PROCESSES OF 4.0 ORDER PROCESSING

Process ID	Process Name	Description
4.1	Receive Order	Receives order details from the customer.
4.2	Verify Customer	Verifies the customer using user information.
4.3	Validate Cart	Validates the cart items, quantity and availability.
4.4	Calculate Total Amount	Calculates total amount including item prices, taxes and delivery charge.
4.5	Generate Order ID	Generates a unique order ID for the order.
4.6	Store Order	Stores the order in the Orders database.
4.7	Send Order to Restaurant	Sends order information to the restaurant.
4.8	Notify Customer	Notifies the customer about the delivery status.

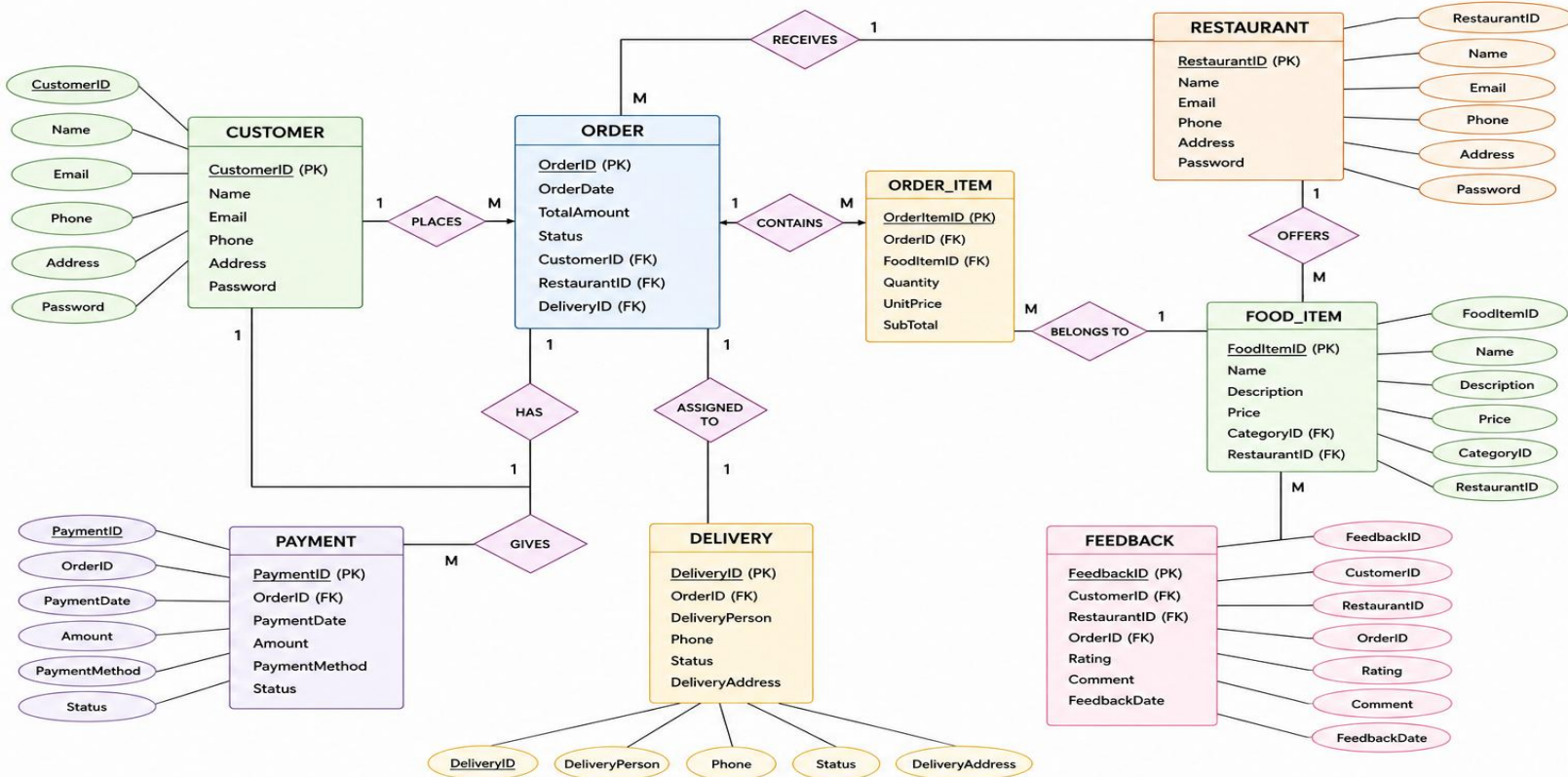
DATA STORES USED

Data Store	Description
D1 Users	Stores user / customer information.
D2 Restaurants	Stores restaurant information.
D3 Food Items	Stores food items, prices and availability.
D4 Cart	Stores customer cart information.
D5 Orders	Stores all order details.
D6 Delivery Records	Stores delivery assignments and status.

(b) Class-based models :

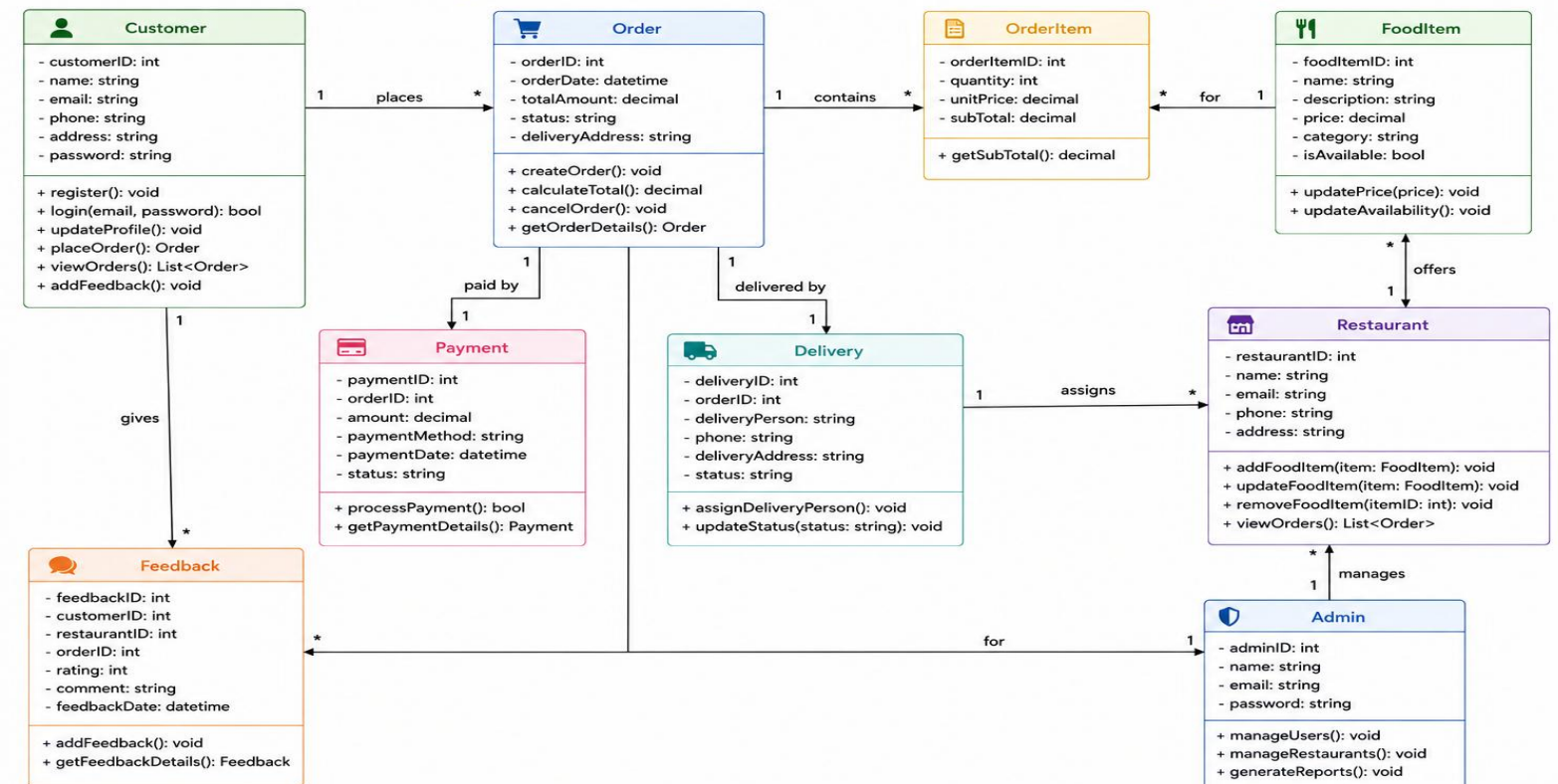
I. ER diagram -

ER Diagram – QuickBite: Online Food Ordering System



I. Class diagram –

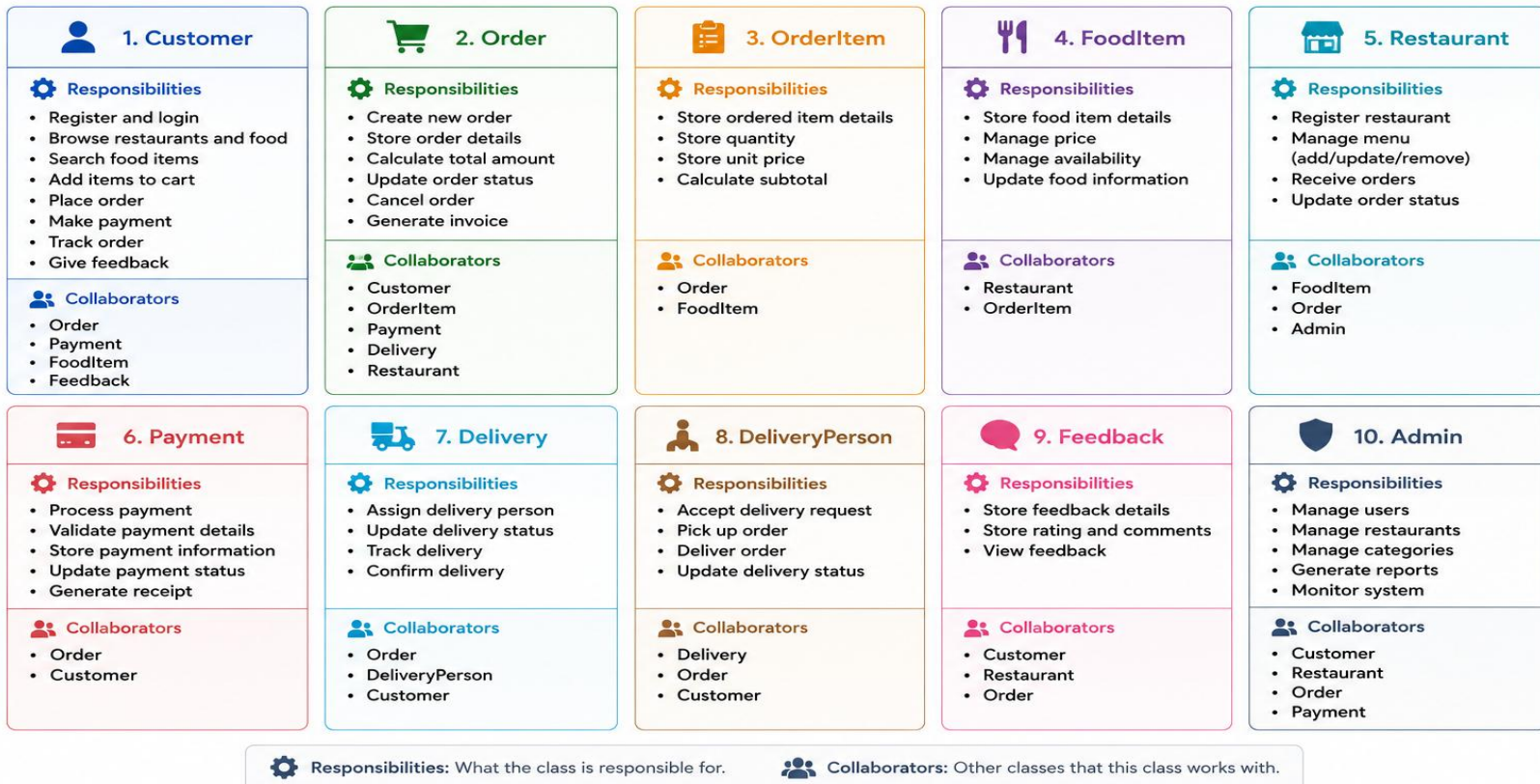
Class Diagram – QuickBite: Online Food Ordering System



III. Class diagram CRC modeling :

CRC Modeling – QuickBite: Online Food Ordering System

CRC = Class, Responsibilities, Collaborators



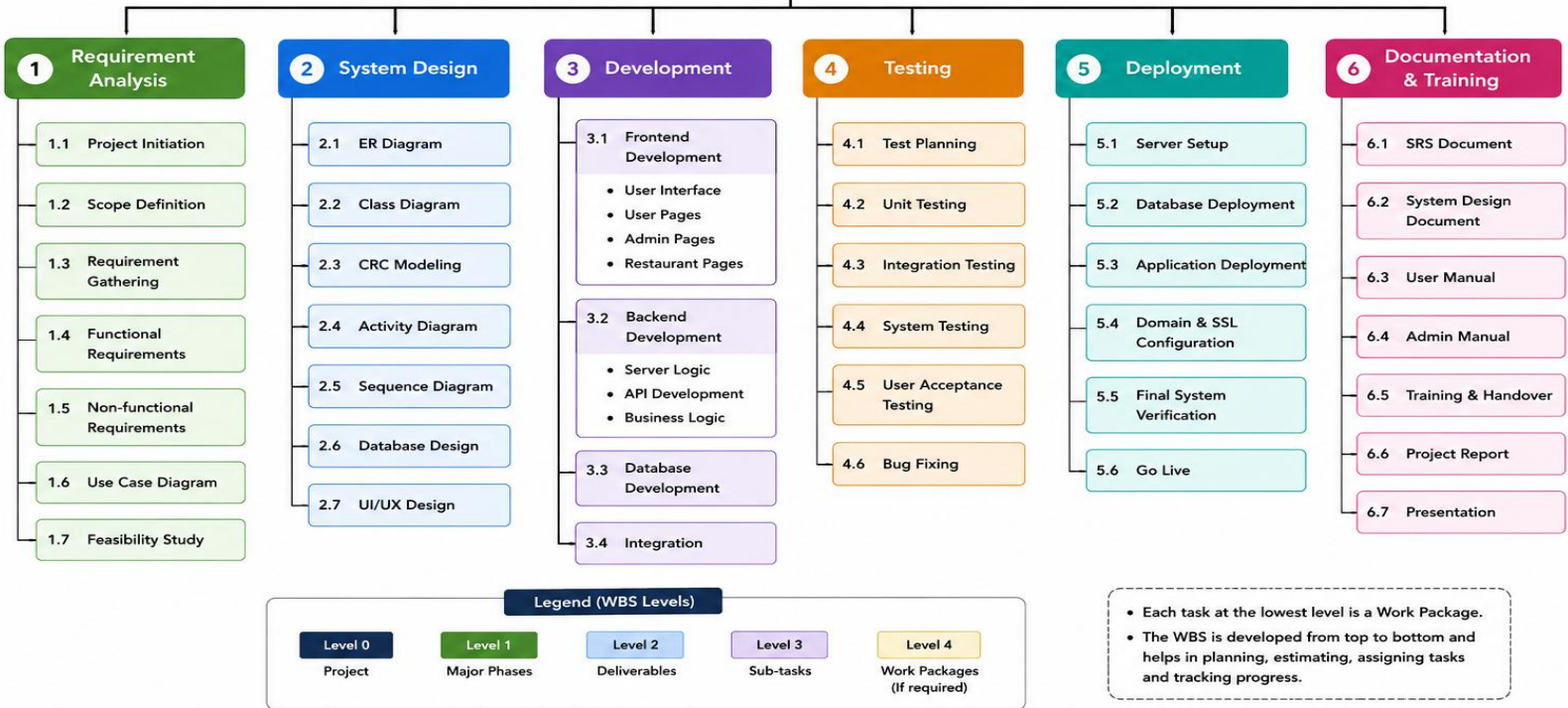
6. Project Planning :

I. WBS –

- The Work Breakdown Structure (WBS) is a hierarchical decomposition of the QuickBite project into smaller and manageable tasks. It helps in organizing project activities, assigning responsibilities, estimating effort, and monitoring project progress effectively.
- The QuickBite project is divided into the following major phases:

WBS (WORK BREAKDOWN STRUCTURE)

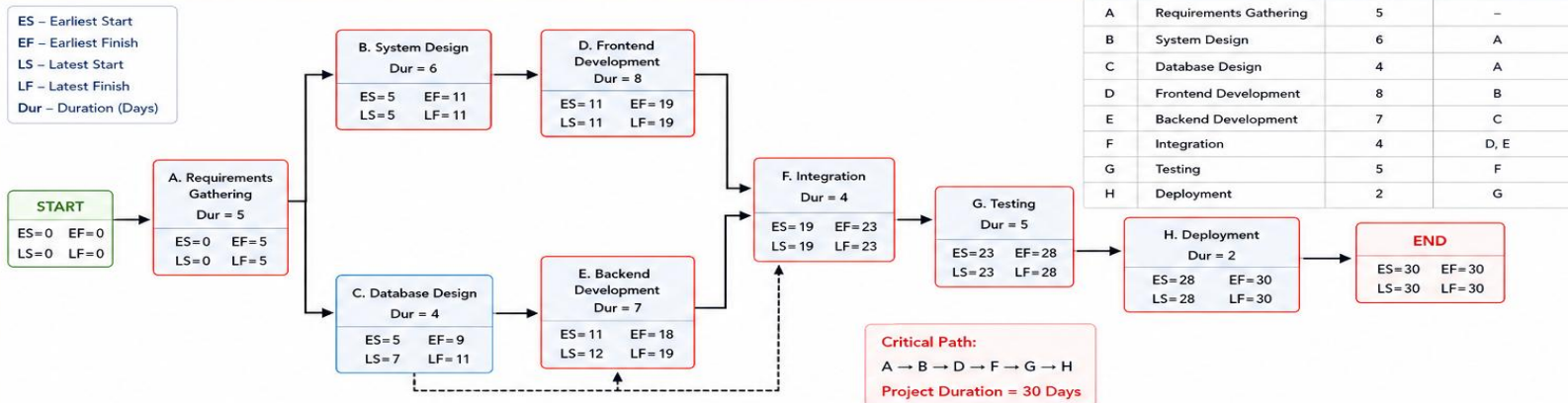
Project: QuickBite – Online Food Ordering System



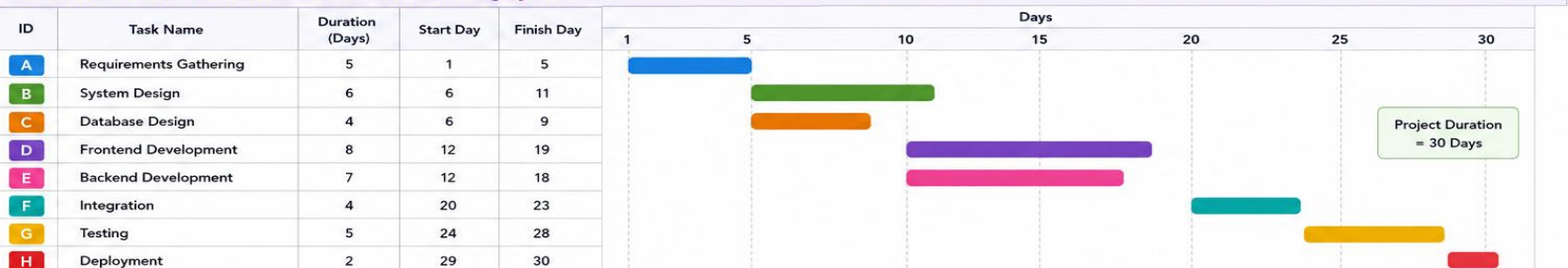
II. Project scheduling: CPM and Gantt chart :

PROJECT SCHEDULING – CPM AND GANTT CHART

A. CPM (Critical Path Method) – QuickBite: Online Food Ordering System



B. GANTT CHART – QuickBite: Online Food Ordering System



- The critical path (red bars) determines the minimum project completion time.
- Any delay in critical path tasks will delay the entire project.

iii) Project Estimation (with Proper Justification)

1. Product Size

QuickBite is a medium-scale web application containing approximately 12–15 database entities and 30+ interfaces/modules, including authentication, menu management, ordering, payment, delivery, and admin functionalities.

2. Effort

Estimated effort is approximately 270 person-hours (34 person-days), covering requirement analysis, design, implementation, testing, and documentation.

3. Schedule

The project is expected to be completed within 14 weeks.

Activity	Duration
Analysis & Design	3 Weeks
Development	6 Weeks
Testing	2 Weeks
Documentation	1 Week
Buffer Time	2 Weeks

4. Cost:

The exact cost has not yet been determined. Since the project uses open-source technologies, the development cost is expected to be minimal, with possible expenses related to hosting, internet usage, and documentation.

7. Risk analysis :

I. Identify risks associated with your project (SWOT analysis)

Risk analysis is conducted to identify the potential strengths, weaknesses, opportunities, and threats associated with the QuickBite project. This analysis helps in understanding possible challenges and developing strategies to minimize project risks and improve overall project success.

RISK ANALYSIS (SWOT ANALYSIS)

Project: QuickBite – Online Food Ordering System

A. SWOT Analysis

 Strengths	 Weaknesses	 Opportunities	 Threats
• User-friendly and easy to use • Online ordering saves time • Multiple payment options • Real-time order tracking • Attractive and modern interface • Centralized management of restaurants and orders	• Limited experience in large scale development • Small development team • Limited budget • Dependence on internet connectivity • Initial version may lack some advanced features	• Growing demand for online food delivery • Integration with more restaurants • Future mobile application development • Use of AI/ML for recommendations • Expansion to wider areas	• Strong competition from existing platforms • Security and privacy risks • Changes in customer preferences • Technical failures or server downtime • Increase in operational costs

B. Risk Identification and Mitigation

Risk ID	Risk Description	Category	Probability	Impact	Risk Level	Mitigation Strategy
R1	Requirement changes during development	Project	Medium	High	Medium	Finalize requirements at early stage and maintain proper documentation.
R2	Delay in completing project tasks	Schedule	High	High	High	Prepare detailed schedule and monitor progress regularly.
R3	Technical difficulties or bugs	Technical	Medium	High	Medium	Use proper tools, follow standards and test modules regularly.
R4	Security vulnerabilities	Security	Medium	High	High	Implement authentication, input validation and secure coding practices.
R5	Database failure or data loss	Technical	Low	High	Medium	Regular database backup and data recovery plan.
R6	User adoption may be low	Business	Low	Medium	Low	Provide better user experience and awareness.
R7	Server downtime or performance issues	Operational	Medium	High	Medium	Use reliable hosting and monitor system performance.

Risk Level Key:

High (H) – Requires immediate attention

Medium (M) – Monitor and control

Low (L) – Acceptable risk

II. RMMM plan :

RMMM Plan – QuickBite: Online Food Ordering System

Risk 1: Delay in Project Completion

1. Mitigation (Avoid Risk)

- ✚ Prepare a detailed project schedule before development.
- ✚ Divide the project into smaller modules.
- ✚ Allocate sufficient time for testing and documentation.

2. Monitoring

- ✚ Monitor weekly progress against the planned schedule.
- ✚ Compare completed tasks with the Gantt chart.
- ✚ Review milestones regularly.

3. Management

- ✚ Reassign tasks if delays occur.
- ✚ Reduce non-essential features if necessary.
- ✚ Request additional time from the supervisor when required.

Risk 2: Requirement Changes

1. Mitigation

- ✦ Gather and finalize requirements at the beginning of the project.
- ✦ Obtain supervisor approval before implementation.

2. Monitoring

- ✦ Review requirement documents periodically.
- ✦ Track requested modifications.

3. Management

- ✦ Update design documents and diagrams accordingly.
 - ✦ Adjust the development schedule if major changes arise.
-

Risk 3: Security Issues

1. Mitigation

- ✦ Use secure authentication methods.
- ✦ Validate user inputs.
- ✦ Encrypt passwords.

2. Monitoring

- ✦ Perform regular testing for vulnerabilities.
- ✦ Monitor login activities and database access.

3. Management

- ✦ Fix security issues immediately.
 - ✦ Backup data regularly.
 - ✦ Implement recovery mechanisms.
-

Risk 4: Database Failure

1. Mitigation

- ✦ Design a normalized database.
- ✦ Maintain backup copies.

2. Monitoring

- ✦ Check database consistency periodically.
- ✦ Monitor query performance.

3. Management

- ✦ Restore data from backups.
 - ✦ Repair corrupted records if necessary.
-